For instance if we draw a sample of size n from a given finite population of size N, then the total number of possible samples is ${}^{N}C_{n} = \frac{N!}{n!(N-n)!} = k$ (say). For each of these k samples we can compute some statistic, $t = t(x_1, x_2, x_3, \ldots x_n)$, in particular the mean $\bar{x}$, the variance S^2, etc., is given below:

Sample Number	Statistic		
	t	$\bar{x}$	s^2
1	t_1	$\bar{x}_1$	s_1^2
2	t_2	$\bar{x}_2$	s_2^2
3	t_3	$\bar{x}_3$	s_3^2
. . .	. . .	. . .	. . .
k	t_k	$\bar{x}_k$	s_k^2

The set of the values of the statistic so obtained, one for each sample, constitutes the sampling distribution of the statistic.

Standard Error

The standard deviation of the sampling distribution of a statistic is known as its Standard Error abbreviated as S.E. The Standard Errors (S.E.) of some of the well-known statistics, for large samples, are given below, where n is the sample size, σ^2 is the population variance.

Sl. No.	Statistic	Standard Error
1.	Sample mean ($\bar{x}$)	$\sigma/\sqrt{n}$
2.	Observed sample proportion (p)	$\sqrt{PQ/n}$
3.	Sample standard deviation (s)	$\sqrt{\sigma^2/2n}$
4.	Sample variance (s^2)	$\sigma^2\sqrt{2/n}$
5.	Sample quartiles	$1.36263\,\sigma/\sqrt{n}$
6.	Sample median	$1.25331\,\sigma/\sqrt{n}$
7.	Sample correlation coefficient (r)	$(1-\rho^2)/\sqrt{n}$

8.1.4 Computing standard error in simple cases

Example 8.6

A server channel monitored for an hour was found to have an estimated mean of 20 transactions transmitted per minute. The variance is known to be 4. Find the standard error.

Solution:

Given $\sigma^2 = 4$ which implies $\sigma = 2$, $n = 1$ hour = 60 min, $\bar{X} = 20$ /min

Standard Error = $\frac{\sigma}{\sqrt{n}} = \frac{2}{\sqrt{60}} = 0.2582$

Example 8.7

Find the sample size for the given standard deviation 10 and the standard error with respect of sample mean is 3.

Solution:

Given $\sigma = 10$, S.E. $\bar{X}$ =3 We know that S.E = $\sigma/\sqrt{n}$

Therefore, $3 = \frac{10}{\sqrt{n}} \Rightarrow \sqrt{n} = \frac{10}{3}$

Taking Squaring on both sides we get

$$n = \left(\frac{10}{3}\right)^2 = \frac{100}{9} = 11.11 \cong 11,$$

The required sample size is 11.

Example 8.8

A die is thrown 9000 times and a throw of 3 or 4 is observed 3240 times. Find the standard error of the proportion for an unbiased die .

Solution :

If the occurrence of 3 or 4 on the die is called a success, then

Sample size = 9000; Number of Success = 3240

Sample proportion = $p = \frac{3240}{9000} = 0.36$

Population proportion (P) = Prob (getting 3 or 4 when a die is thrown)

$$= \frac{1}{6} + \frac{1}{6} = \frac{2}{6} = \frac{1}{3} = 0.3333$$

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